

Year 7 • Chapter 12 • Ratios, rates and time: answers (teacher copy)

12.01 Ratios (booklet p. 3)

WE DO examples

- Ex 1: 3 : 5
- Ex 2: 4 : 6
- Ex 3: 38 : 17
- Ex 4: 17 : 55
- Ex 5: 2 : 1
- Ex 6: No: the order matters

Set A

- 1 : 5
- 5 : 3
- 4 : 5
- 7 : 5
- 2 : 4
- 3 : 7

Set B

- 13 : 12
- 12 : 13
- 13 : 25
- 12 : 25
- 1 : 25
- 25 : 1

Set C

- 7 : 4
- 4 : 11
- 9 : 5
- 5 : 16
- 3 : 2
- 4 : 9

Set D

- 1/5
- 8
- 3 shaded, 2 unshaded in each 5 (6 : 4)
- yes, e.g. 2 : 3 as $\frac{2}{3}$ of the second amount

Extension

- $\frac{3}{10}$
- 30%

Exit ticket

- Level 1: 6 : 5
- Level 2: 5 : 11
- Level 3: 6 : 7

12.02 Equivalent and simplified ratios

(booklet p. 7)

WE DO examples

- Ex 1: 8 : 12
- Ex 2: 6 : 9 : 12
- Ex 3: 4 : 7
- Ex 4: 3 : 1 : 4
- Ex 5: 2 : 1
- Ex 6: 1 : 3

Set A

- 12
- 10
- 15
- 6
- 9, 12
- 3

Set B

- 4 : 7
- 1 : 3
- 1 : 6
- 5 : 9
- 3 : 1 : 4
- 1 : 5 : 20

Set C

- 2 : 1
- 10 : 1
- 5 : 1
- 1 : 20
- 1 : 5
- 1 : 2

Set D

- 1 : 3
- 4 : 7
- 48 : 5
- 3 : 5

Extension

- 6 : 5
- 1 : 3 : 6

Exit ticket

- Level 1: 21
- Level 2: 2 : 5
- Level 3: 3 : 20

12.03 Ratio problems (booklet p. 11)

WE DO examples

- Ex 1: 100
- Ex 2: \$45 000
- Ex 3: \$20 and \$40
- Ex 4: \$200 and \$250
- Ex 5: 1.5 km
- Ex 6: 12 kg

Set A

- 16
- 12
- 10
- 48
- 20
- 45

Set B

- \$20, \$40
- \$40, \$60
- \$20, \$25
- 45, 27
- \$200, \$800
- 8, 16, 24 kg

Set C

1. 100
2. \$45 000
3. 15 L
4. \$200, \$250
5. 1.5 km
6. 12 kg

Set D

1. 320
2. Sam \$36, Jo \$60
3. 750 g
4. 30 cm by 45 cm

Extension

1. 12
2. A \$28, B \$14, C \$42

Exit ticket

1. Level 1: 35
2. Level 2: \$30 and \$50
3. Level 3: 1 L

12.04 Rates (booklet p. 15)

WE DO examples

1. Ex 1: 3000 voters/politician
2. Ex 2: 200 sheep/ha
3. Ex 3: \$0.80/ticket (or 1.25 tickets/\$)
4. Ex 4: 270 stitches/ball
5. Ex 5: 80 km/h
6. Ex 6: 1500 m/min

Set A

1. 3000 voters/politician
2. 200 sheep/ha
3. \$0.80/ticket
4. 270 stitches/ball
5. \$15/h
6. 80 km/h

Set B

1. \$5/kg
2. 15 L/min
3. 90 words/min
4. 1.2 beats/s
5. \$0.90/L
6. 14 km/L

Set C

1. 80 km/h
2. 5 L/min
3. \$22/h
4. 800 km/h
5. 25 pages/min
6. 5 m/s

Set D

1. 1500 m/min
2. 18 km/h
3. 30 c/min
4. 20 m/s (= 72 km/h)

Extension

1. 54 km/h
2. 10.8 m/h

Exit ticket

1. Level 1: \$9/kg
2. Level 2: 75 km/h
3. Level 3: 1000 m/min

12.05 Best buys (booklet p. 19)

WE DO examples

1. Ex 1: \$1.30
2. Ex 2: \$1.11; Chow
3. Ex 3: Dairy Farm (\$2.85/L vs \$3.12/L)
4. Ex 4: 3 L (\$2.20/L vs \$2.30/L)
5. Ex 5: e.g. it may go off or they cannot store it
6. Ex 6: 5 for \$6.50 (\$1.30 vs \$1.40)

Set A

1. \$0.60
2. \$0.65
3. \$3.50/kg
4. \$1.70/L
5. \$0.75
6. \$1.40/m

Set B

1. \$1.30
2. \$1.11
3. \$1.20
4. \$0.85
5. \$0.80
6. \$0.95

Set C

1. Chow
2. 1 L (\$2.85/L vs \$3.12/L)
3. 3 L
4. 800 g (\$0.70 vs \$0.75 per 100 g)
5. 10 rolls (\$0.75 vs \$0.80)
6. 1.25 L (\$3.60/L vs \$4/L)

Set D

1. 5 kg (\$2.10/kg)
2. 5 c per 100 g
3. no: waste, storage, use-by dates
4. own answer

Extension

1. 1 L: \$1.35 vs \$1.50 per 100 mL, 15 c cheaper
2. \$3.00

Exit ticket

1. Level 1: \$4.20/kg
2. Level 2: 2 L (\$2.70/L)
3. Level 3: 1 kg (\$0.90/100 g)

12.06 Rate problems (booklet p. 23)

WE DO examples

1. Ex 1: \$853.10
2. Ex 2: 36.3 L
3. Ex 3: 7.5 ha
4. Ex 4: 200 km
5. Ex 5: 3 h
6. Ex 6: 17.28 L

Set A

1. \$853.10
2. 200 km
3. \$14
4. 120 L
5. 720 words
6. \$71.60

Set B

1. 36.3 L
2. 7.5 ha
3. 3 h
4. 15 min
5. 12 h
6. 25 min

Set C

1. 17.28 L
2. \$417.50
3. 18 km
4. 3.36 kg
5. 103 680
6. 27 L

Set D

1. 30 min
2. \$66.60
3. 10 min
4. 100 m in 12.5 s (8 m/s vs 6.67 m/s)

Extension

1. 360 km
2. after 2 hours

Exit ticket

1. Level 1: \$147
2. Level 2: 3 h
3. Level 3: 31.25 L

12.07 Travel graphs (booklet p. 27)**WE DO examples**

1. Ex 1: 22 km
2. Ex 2: 2
3. Ex 3: 3 p.m.
4. Ex 4: 1 p.m. to 2 p.m.
5. Ex 5: 4 km/h
6. Ex 6: faster

Set A

1. 22 km
2. 2
3. 3 p.m.
4. 1 p.m. to 2 p.m. (5 km/h)

Set B

1. 300 km
2. 1 hour
3. 80 km/h
4. 40 km/h

Set C

1. travel graph

Set D

1. stopped (not moving)
2. returning to the start
3. e.g. drove 160 km, stopped for lunch for 1 h, then drove on
4. 60 km/h

Extension

1. graph
2. 10 a.m.

Exit ticket

1. Level 1: stopped
2. Level 2: 15 km/h
3. Level 3: 3 km

12.08 Time (booklet p. 31)**WE DO examples**

1. Ex 1: 10 h
2. Ex 2: 3 h; 12 min
3. Ex 3: 5 h 16 min
4. Ex 4: 3 min 28 s
5. Ex 5: 10 h 5 min
6. Ex 6: 29 min

Set A

1. 10 h
2. 3 h
3. 4 h
4. 8 h
5. 2 h
6. 12 h

Set B

1. 5 h 16 min
2. 3 min 28 s
3. 150 min
4. 195 min
5. 24 min
6. 1440 min

Set C

1. 10 h 5 min
2. 4 h 42 min
3. 2 h 9 min
4. 29 min
5. 7 h
6. 1 h 30 min

Set D

1. 9:53 p.m.
2. 10 min 55 s
3. 15 s
4. 2 h 51 min

Extension

1. age $\times$ 8760 (approx.)
2. 11 days 13 h 46 min 40 s

Exit ticket

1. Level 1: 2 h 30 min
2. Level 2: 4 h 15 min
3. Level 3: 2 h 25 min

12.09 12-hour and 24-hour time (booklet p. 35)**WE DO examples**

1. Ex 1: 3:50 p.m.
2. Ex 2: 1:48 a.m.
3. Ex 3: 12:30 p.m.
4. Ex 4: 14:10
5. Ex 5: 00:19
6. Ex 6: no confusion between a.m. and p.m.

Set A

1. 1:48 a.m.
2. 6:24 a.m.
3. 3:50 p.m.
4. 12:30 p.m.
5. 11:05 p.m.
6. 12:40 a.m.

Set B

1. 14:10
2. 05:33
3. 00:19
4. 13:50
5. 23:45
6. 12:05

Set C

1. 3:45 p.m.
2. 35 min
3. the 15:10 bus (arrives 15:31)
4. every 30 minutes

Set D

1. 02:15, 2:15 a.m.
2. 7 h 45 min
3. 23:40
4. 7:15 p.m. (19:15)

Extension

1. 05:35
2. 515 min

Exit ticket

1. Level 1: 5:25 p.m.
2. Level 2: 20:05
3. Level 3: 00:45

12.10 Time differences (booklet p. 39)

WE DO examples

1. Ex 1: 4 h 20 min
2. Ex 2: 7 h 35 min
3. Ex 3: 6 h 30 min
4. Ex 4: 30 min
5. Ex 5: 19 years 2 months 13 days
6. Ex 6: 18 days

Set A

1. 4 h 20 min
2. 6 h 55 min
3. 6 h 30 min
4. 2 h 25 min
5. 5 h 35 min
6. 2 h 25 min

Set B

1. 7 h 35 min
2. 30 min
3. 11 h 55 min
4. 7 h 45 min
5. 7 h 40 min
6. 5 h 45 min

Set C

1. 18 days
2. 45 days
3. 8 weeks 4 days
4. 19 years 2 months 13 days
5. own answer
6. 1 day 6 h 35 min

Set D

1. 4 h 55 min
2. 33 days
3. 8 years 10 months
4. \$12.80 (4 part hours)

Extension

1. 10:00 a.m.
2. 07:00 the next day (Sydney)

Exit ticket

1. Level 1: 2 h 45 min
2. Level 2: 2 h 40 min
3. Level 3: 15 days

12.11 Timetables (booklet p. 43)

WE DO examples

1. Ex 1: 8:33 a.m.
2. Ex 2: 2 h 16 min
3. Ex 3: 7:40 a.m.
4. Ex 4: 1 h 13 min
5. Ex 5: 27 min
6. Ex 6: no confusion between a.m. and p.m.

Set A

1. 8:21 a.m.
2. 2 h 16 min
3. 9:25 a.m.
4. 1 h 23 min

Set B

1. 8:33 a.m.
2. 2 h 16 min
3. 7:40 a.m.
4. 1 h 13 min
5. 22:01
6. 1 h 35 min

Set C

1. 27 min
2. 16:18, reaching the Beach at 16:45
3. the 15:40 bus (Shops 16:01); the 16:10 bus reaches the Shops at 16:31, too late
4. 4:10, 4:18, 4:31, 4:45 p.m.

Set D

1. 06:50 → 07:15, 07:30 → 07:55, 08:10 → 08:35, 08:50 → 09:15, 09:30 → 09:55
2. own timetable

Extension

1. 8:33 a.m.
2. 10:08 a.m.

Exit ticket

1. Level 1: 10:30 a.m.
2. Level 2: 21 min
3. Level 3: the 6:05 a.m. train